

Chapter 2. Literature review and formulation of research problem

Research process:

-The research process is a systematic way of collecting, analyzing, and interpreting information to answer questions or solve problems. It ensures that the findings are valid, reliable.

-The research process involves a systematic series of steps, beginning with identifying a research problem or topic and conducting a literature review to understand existing knowledge. Researchers then formulate research questions and hypotheses, and develop a research design, including sampling and data collection methods. The collected data is then analyzed, leading to the drawing of conclusions and interpretation of findings. Finally, the entire process culminates in writing a research report to communicate the study's outcomes.

- 1. Identifying the Research Problem

Recognize an issue, question, or gap in knowledge.

Clearly define what you want to study.

2. Review of Literature

Study previous research work related to your topic.

Helps understand what is already known and where the gap exists.

Prevents duplication of work.

3. Defining Objectives / Hypothesis

Set clear goals or aims of the research.

Formulate hypotheses (if applicable) that can be tested.

4. Research Design

Plan how the research will be conducted.

Decide the type of research: qualitative, quantitative, or mixed.

Select research methods: survey, experiment, case study, interview, etc.

5. Sampling Design

Select who or what will be studied (population and sample).

Decide sampling techniques (random, stratified, purposive, etc.).

6. Data Collection

Gather information using appropriate tools:

Primary data → questionnaires, interviews, observations, experiments.

Secondary data → books, journals, reports, online databases.

7. Data Processing and Analysis

Organize collected data (editing, coding, tabulating).

Apply statistical or qualitative analysis techniques.

Interpret results logically.

8. Testing Hypothesis (if any)

Use analysis to accept or reject hypotheses.

Compare results with expectations.

9. Interpretation of Results

Explain findings in relation to objectives or research questions.

Highlight implications, significance, and contributions.

10. Report Writing and Presentation

Present research in a structured format: introduction, methodology, results, discussion, and conclusion.

Share findings with others (journals, conferences, presentations).

11. Conclusion and Recommendations

Summarize key findings.

Suggest practical applications or future research directions.

12. Write the Research Report:

Document the entire research process, including the problem statement, methodology, results, and conclusions, to share the findings with others.

Reviewing the literature: purpose of a literature review

-A literature review means reading and studying previous research, books, and articles on the topic you are working on.

-👉 In short:

A literature review is done to learn from past research, find gaps, and give a solid base to your own study.

-A literature review is the process of reading, analyzing, and summarizing existing research, books, articles, and studies related to your research topic.

-* Purpose of a Literature Review :

1. Understand what is already known

→ It tells you what other researchers have already found.

2. Find gaps in knowledge

→ Helps you see what is missing or unanswered, so you can work on that.

3. Avoid repeating old work

→ Saves time by not doing the same research again.

4. Get new ideas

→ Reading others' work can inspire your own approach or methods.

5.Build a strong foundation

→ Gives background information and context for your study.

6.Support your research

→ Shows that your study is connected to previous knowledge, and proves why it is important.

7.Help in choosing methods

→ You can learn what methods worked for others and use or improve them.

8.Avoid Duplication of Effort:

Reviewing existing work prevents you from repeating research that has already been done.

***Literature resources**

Literature:

- When we do a literature review, we need to collect information from reliable literature resources.

-Literature resources are places where you find information for your research — like books, journals, databases, reports, and theses.

-When we do research, we need to collect information from literature resources. These are the places or materials where knowledge is already written and stored.

-Literature resources encompass a wide range of materials used for research, study, and enjoyment of literature, including books, journals, databases, and online platforms. These resources can be categorized into primary sources (original literary works) and secondary sources (criticism, analysis, and interpretations).

-*Types of Literature Resources:

 In short:

Primary = Original research

Secondary = Summary of research

Tertiary = Tools to find research

1. Primary Sources (Original Information)

These are first-hand materials — created by the person who actually did the research or experienced the event.

Examples:

Research journal articles (original experiments, surveys, case studies)

Conference papers (presented by researchers)

Official reports, patents, government publications

Historical records, diaries, speeches

👉 Why useful? Because they give fresh, original data.

2. Secondary Sources (Summarized Information)

These are based on primary sources, but already explained, analyzed, or summarized by others.

Examples:

Review articles (summaries of many research papers)

Books and textbooks (explaining theories and concepts)

Magazines, newspapers (summaries, not original studies)

👉 Why useful? Because they save time and give you an overview.

3. Tertiary Sources (Guides to Information)

These are tools that help you find primary and secondary sources.

Examples:

Indexes (lists of research titles)

Abstracts (short summaries of research papers)

Databases (Google Scholar, PubMed, IEEE Xplore, Scopus, JSTOR)

Bibliographies (list of references at the end of a book or article)

👉 Why useful? Because they guide you to the right materials.

The internet & a literature view

- A literature review means reading what others have already written about your topic.**
- The internet helps a lot because it gives you books, research papers, and articles online.**
- The internet makes a literature review easy and fast, but you must choose reliable sources only.**

-👉 In short:

The internet makes a literature review faster, easier, and broader, but researchers must be careful to use authentic and reliable sources only.

-The internet has become one of the most important tools for doing a literature review because it gives quick and wide access to information.

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***How the Internet Helps in Literature Review**

1.Easy Access to Research

Online journals, e-books, articles, and reports are available.

Example: Google Scholar, ResearchGate, PubMed, JSTOR.

2.Latest Information

Internet gives access to the most recent studies faster than printed books.

3.Wide Variety of Sources

From primary sources (original research papers) to secondary (reviews, summaries) and tertiary (databases, indexes).

4.Search Tools

Search engines and databases make it easy to find relevant studies using keywords.

5.Global Knowledge

Researchers from all over the world can share work online, so you get international perspectives.

6.Efficiency and Speed:

Search engines and online databases allow researchers to quickly locate relevant literature, saving time and effort compared to manual searches.

*****ΔBe Careful**

Not everything on the internet is true.

Always use trusted websites (like university sites, online journals, databases).

Avoid random blogs or Wikipedia for serious research.**

The internet & resources strategies & methods

***The internet & research strategies & methods**

-The internet is now a very important tool for doing research. It helps at every stage — from finding a topic to collecting data and presenting results.

-The internet supports research strategies by helping in planning, finding, and reviewing information.

-It also supports research methods by providing tools for data collection (surveys, interviews, analytics) and analysis.

1. Role of Internet in Research Strategies:

A research strategy means the overall plan to carry out research. The internet helps by:

Finding a topic → Searching trending issues and gaps.

Accessing literature → Using online databases (Google Scholar, PubMed, IEEE, JSTOR).

Choosing methods → Reading how other researchers studied similar topics.

Collecting data → Online surveys, forms, interviews (Zoom, Google Forms, SurveyMonkey).

Collaboration → Connecting with other researchers worldwide.

2. Role of Internet in Research Methods

Research methods are the actual tools and techniques to collect and analyze data.

The internet provides:

Quantitative Methods (numbers/data):

Online interviews (Zoom, Skype, Google Meet)

Focus groups (discussion groups online)

Social media analysis (Twitter, Facebook, Instagram data)

***Advantages of Using the Internet in Research**

Fast and easy access to huge information.

Low cost (many resources are free).

Wider reach (can collect data from people worldwide).

Up-to-date information.

***Limitations**

-Not all information is reliable.

-Risk of plagiarism (copy-paste without credit).

-Some journals/websites require paid access.

Conducting & evaluating literature reviews

👉 In short:

Conducting a literature review = searching, reading, organizing, and writing about past studies.

Evaluating a literature review = checking if it is complete, critical, clear, and relevant.

1. Conducting a Literature Review (How to do it)

This means step-by-step process of preparing a literature review.

I] .Choose your topic → Be clear about your research question.

II].Search for literature → Use books, journals, databases, and the internet (Google Scholar, PubMed, JSTOR, IEEE).

III] Collect sources → Gather research papers, articles, theses, reports.

IV] Read and take notes → Summarize the important ideas, methods, and results.

V].Organize the information → Group similar studies together (themes, methods, findings).

VI].Identify gaps → See what is missing in existing research.

VII].Write the review → Explain what is already known, what is missing, and how your research will add to it.

2. Evaluating a Literature Review (Checking its Quality)

After writing or reading a literature review, you must evaluate it to see if it is good.

Good Literature Review should:

Be relevant → Covers studies directly related to your topic.

Be comprehensive → Includes both old and recent research.

Be critical → Not just summary, but analysis (strengths & weaknesses of past work).

Be organized → Clear flow (themes, chronology, or methods).

Formulation of research problem

***What is a Research Problem?**

A research problem is the question or issue that you want to study in your research.

👉 It is the starting point of any research.

***Formulation of a Research Problem:**

This means turning a general idea into a clear, specific, and researchable problem.

1.what is research problem:

- A research problem is the question, difficulty, or issue that a researcher wants to study and solve.

-A research problem is simply a question or issue that you want to find an answer to through research.

-It is like the “puzzle” you want to solve.

-It is the starting point of any research because without a problem, there is no reason to do research.

2.Se necessity of defining the problem

-Defining a research problem clearly is very important because it guides the entire research process.

-Defining the research problem = knowing exactly what question you want to answer, so your research is clear, focused, and useful.

- *Reasons / Necessity:

1.Gives Direction

A clear problem tells you what to study and keeps the research focused.

2.Helps in Planning

You can decide methods, tools, and data to use based on the problem.

3.Avoids Wasting Time

Without a defined problem, you may collect irrelevant information or do unnecessary work.

4.Clarifies Objectives

Makes it easier to set specific goals and questions for your study.

5.Ensures Feasibility

Helps check if the study is possible with available resources and time.

6.Helps in Analysis

When the problem is clear, it is easier to analyze data and interpret results.

7.Communicates Importance

A well-defined problem shows others why your research is useful and significant.

3. Technique involved in defining a problem

- Defining a research problem means making a broad idea clear, specific, and researchable. Here are some techniques:

-1. Literature Survey / Review

Read books, journals, articles, and reports on your topic.

Helps you understand what has already been studied and what is missing.

2. Consulting Experts

Talk to teachers, professors, or professionals in the field.

They can guide you about important issues or gaps.

3. Personal Observation

Look around in real life and observe problems or trends.

Example: Watching how students use mobile phones in class.

4. Experience / Practical Knowledge

Use your own experience to identify issues.

Example: A teacher noticing students struggle with online learning.

4.selecting the problem

- Selecting a problem means choosing a question or issue that you want to study in your research.

-It is an important step because a good problem makes research easier and meaningful.

-Selection of research problem depends on several factors such as researcher's knowledge, skills, interest, expertise, motivation & creativity with respect to the subject of inquiry.

***Qualities of a Good Research Problem**

Clear → Easy to understand

Specific → Not too broad

Researchable → Can be studied with data

Significant → Important for knowledge or society

Feasible → Can be done with available resources

5. Necessity of Defining the problem

- Defining a research problem clearly is very important because it guides the entire research process and ensures the study is meaningful and effective.

- Defining the research problem = knowing exactly what question you want to answer, so your research is clear, focused, organized, and meaningful.

-* necessity

1.Gives Direction to Research:

**A clearly defined problem tells you what to study.
Prevents confusion and keeps the research focused.**

2.Helps in Planning:

**Determines what methods, tools, and data to use.
Makes the research process organized.**

3.Avoids Wasting Time and Effort:

Without a defined problem, you may collect irrelevant data or work on unnecessary tasks.

4.Clarifies Objectives:

**Helps set specific goals and research questions.
Makes it easier to know what you want to find out.**

5.Ensures Feasibility:

Helps check if the research can be done with available time, money, and resources.

6.Helps in Data Analysis and Interpretation:

When the problem is clear, it is easier to analyze results and draw conclusions.

7.Shows Significance:

A well-defined problem communicates why your research is important and useful for knowledge or society.

6. Technique involved in defining a problem

- Defining a research problem means turning a broad idea into a clear, specific, and researchable question.**
- Techniques to define a research problem = reading, asking experts, observing, using experience, brainstorming, and checking feasibility.**
- Techniques to define a research problem are ways to make a research question clear and specific.**
- You can read books and articles, ask experts, observe real-life situations, use your experience, brainstorm ideas, and check feasibility.**
- These steps help you choose a problem that is important, clear, and possible to study.**

*** Techniques:**

1. Literature Survey / Review

Read books, journals, articles, and reports on your topic.

Helps you understand what is already known and find gaps.

2. Consulting Experts:

Talk to teachers, professors, or professionals in the field.

They can guide you about important and relevant issues.

3. Observation:

Watch real-life situations carefully to find problems.

Example: Observing how students use mobile phones during studies.

4. Personal Experience:

Use your own experience or knowledge to identify issues.

Example: A teacher noticing students struggling with online learning.

5. Brainstorming:

Make a list of possible problems and discuss with peers or mentors.

Helps in narrowing down the focus.

6. Feasibility Check:

Ask yourself:

Can I collect data for this problem?

Do I have enough time, money, and resources?

Ensures the problem is practical to study.